

STAT 548 PhD Qualifying Course Papers

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Choosing a paper

In this document, I provide a list of suggested papers and projects I am interested in supervising. However, I am open to discussing papers and projects from outside this list that relate to my research interests. I am broadly interested in methodology for the statistical analysis of high-dimensional biological data. In particular, I develop and apply methods for inference of the genetic/epigenetic basis of disease, as well as mechanisms of gene regulation, from sequencing data (e.g. from DNA, RNA, or epigenome). I am also interested in multiple testing problems in high-dimensional biology.

Report¹

Summary (max 4 pages): The first section of the report should provide a summary of the problem the paper addresses in the context of previous work, limitations of previous work, the solution technique, important results, why they are important, and limitations of the paper. The paper you select may have multiple contribution areas (theory, modeling, computation, biological insights), and the summary should reflect that. The goal of this portion of the report is to show that you can take a complex body of work (one paper and earlier relevant work), digest it, and present a concise summary of the important points.

Mini Project (no page limit): The remainder of the report will be devoted to a paper-specific mini project. This mini project should build upon the paper in some way. This could involve identifying a problem with the method and proposing an improvement, or extending the method for application in a different context. Your approach might use simulation studies and/or publicly available data. Some ideas for suggested projects are included in the list of papers below (you should feel free to stray from these ideas). Please schedule a meeting with me to discuss these and/or other ideas you may have. Note that your grade will not be affected by how good the results look, whether your approach improves on past work, or whether you achieve the initial goal of the project. Your grade for this section will instead be based upon how you approach the problem and how you evaluate and communicate the results².

Project organization

All results in your report should be reproducible. Reproducible research requires that your project is organized and well-documented. The ultimate goal is that, given your code and report, another

¹Adapted from: [Trevor Campbell](#) and [Marie Auger-Méthé](#)

²See also <https://www.stat.ubc.ca/phd-qualifying-course>

researcher (or your future self!) can and obtain and comprehend the same results.

All code used in this project should be housed in a GitHub repository (for which you should add me as a collaborator before submitting the final report). You are encouraged to use git to commit regularly. This helps me keep track of your progress, but also serves as a backup in case you need to revert to an earlier version.

The directory structure should be organized to separate any raw data from derived results. For example, here is a suggested directory structure to organize your project files³:

```
/README.md      <- The top-level README with summary of project
/data
  /raw           <- The original, unchanged data
  /interim       <- Intermediate data that has been transformed
  /processed     <- The final processed data sets for downstream analysis
/references      <- Articles and manuals used for reference
/report         <- Latex files and generated PDF of report
/figures        <- Generated graphics and figures to be used in report
/src            <- Source code: subdirectories will vary by project type/scope
  /01_data       <- Source code to obtain / generate data
  /02_process    <- Source code to process data for downstream analysis
  /03_analysis   <- Source code for downstream analysis
  /04_report     <- Source code to generate final figures/tables in report
```

Your report should be written in Latex. Please make sure to submit all necessary files (.tex, .bib, any style files) necessary for me to compile, as well as the pdf.

Resources on reproducible research

- Karl Broman's guide to reproducible research: <https://kbroman.org/steps2rr/>
- Jenny Bryan's git manual (for those new to Git/GitHub): <https://happygitwithr.com>

Available papers (last updated September 10, 2026)

Please reach out if you have trouble accessing any of these papers, and I can send you a PDF.

1. Generating control samples using deep learning

Paper: Prada-Luengo, I, Schuster, V, Liang, Y, Terkelsen, T, Sora, V, Krogh, A. N-of-one differential gene expression without control samples using a deep generative model. *Genome Biology*. 24:263, 2023.

Link: <https://genomebiology.biomedcentral.com/articles/10.1186/s13059-023-03104-7>

Mini Project idea: Empirically assess the properties of the proposed DGD test under the null (empirical rejection rate, distribution of p-values).

2. Statistical deconvolution of cell types from epigenetic signals of cell-free DNA

Paper: Caggiano, C, Celona, B, Garton, F et al. Comprehensive cell type decomposition of circulating cell-free DNA with CelFiE. *Nature Communications* 12, 2717, 2021.

Link: <https://doi.org/10.1038/s41467-021-22901-x>

³Adapted from <https://drivendata.github.io/cookiecutter-data-science/#directory-structure>

Mini Project idea: Evaluate robustness of the proposed method to mis-specification of the reference panel (i.e. when one or more cell types are present that are not represented in the reference panel).

3. **Measures of reliability in DNA methylation arrays**

Paper: Nazarenko, T, Vavourakis, CD, Jones, A, Evans, I, Schreiberhuber, L, Kastner, C, et al. (2023). Technical and biological sources of unreliability of Infinium probes on Illumina Methylation microarrays. *Clinical Epigenetics*. 16:131, 2024.

Link: <https://link.springer.com/article/10.1186/s13148-024-01739-2>

Mini Project idea: State any assumptions made by their proposed resampling technique for the estimation of the ‘unreliability score’ (e.g. homoskedasticity across m , independence of M/U draws). Extract the negative control probe intensity values from a publicly available data file and assess these assumptions.

4. **Inference of differential methylation for sequencing data**

Paper: Park Y, Wu H. Differential methylation analysis for BS-seq data under general experimental design. *Bioinformatics*. 2016 May 15;32(10):1446-53.

Link: <https://academic.oup.com/bioinformatics/article/32/10/1446/1743267>

Mini Project idea: How (if at all) is type I error controlled for differentially methylated region (DMR) detection? Use simulation to evaluate the False Discovery Rate (FDR) for DMR detection using DSS.

5. **Covariate-weighted False Discovery Rate control**

Paper: Ignatiadis N, Klaus B, Zaugg JB, Huber W. Data-driven hypothesis weighting increases detection power in genome-scale multiple testing. *Nature methods*. 2016 Jul;13(7):577.

Link: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4930141/>

Mini Project ideas: (1) Evaluate performance relative to number of tests - suggest rationale for any variation seen and discuss possible improvements. (2) Suggest possible extensions to handle multiple covariates.